Three stages of temporal interference

Draft23: referee revision approved on 24 October 2026

We develop a mesoscale model of temporal-interference (TI) stimulation with three stages in series: membrane access, nonlinear detection of the difference frequency absent from the field, and circuit amplification. We represent access by an effective membrane time and analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire (QIF) neurons, and its spiking network. We compare spatial maps in one head model and quantify conditional kHz access budgets. For weak fields, a static quadratic detector generates a beat proportional to the carrier-polarization product and a signed, direct-current-like (DC-like) steady source proportional to their summed squares. Transcranial alternating-current (tACS) and direct-current (tDCS) stimulation sources are linear to leading order. At the pinned Jansen–Rit operating point, 0.1–0.3 mV per carrier gives TI sources approximately 2–5% of direct sources at equal per-carrier and direct-input polarizations. Dynamic QIF detection crosses from static curvature to positive-slope rectification at fast carriers. Changing the current ratio moves the hippocampal envelope and mean-square centroids oppositely but only rescales the beat map at fixed state. Near a Hopf bifurcation, resonant gain grows inversely with modal damping and is shared with tACS. In the pinned Jansen–Rit column, coupling changes gain independently of detection; amplification appears mainly in beat-locked population-rate modulation, with little mean-rate change. At stronger inputs, the steady source alone drives the circuit across its gamma threshold, as does an unmodulated carrier matched in root-mean-square (RMS) input. Weak-field beat dose follows the polarization product rather than the envelope minimum. The steady source can act at depth; its sign depends on detector state and dynamics, not carrier polarity. For uniform scalar coupling and detection in homogeneous conductivity, its magnitude has no strict interior maximum. Polarization measurements and amplitude and frequency sweeps test the stages; physiological kHz coupling remains uncalibrated.

The revision separates mathematical results, model demonstrations, and physiological hypotheses. Giulio approved the conditional access analysis, bounded theta-neuron pilot, reproducibility work, and a further reduction in length. The abstract above is the current shared text and is revised with the results. The full PING system and access evidence table remain in the article. The editing baseline is draft22 (`69c106b`); draft20 (`e20c047`) remains the reading baseline. No new anatomical cohort or biological experiment is part of this revision.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Abstract and Introduction	A two-tone field has no difference-frequency line. Access creates local polarization; detection generates a beat and steady source; amplification determines the response. Retain the cellular and network literature, weak-perturbation scope, and single-subject FEM scope.	The reader learns the physical problem and the advances beyond existing mixing and network models: joint sources, independent gain, and the dynamic crossover.
Theory: population curve and field coupling	Keep the JR sigmoid, synaptic response, and direct polarization. Derive the quadratic cross term and constant term together. Define and number the dynamic coefficient; retain its weak-input, memory, and nonzero-coefficient conditions.	The rate curve first explains what mixes the carriers. The NMM1 operator notation moves to the model appendix; it is not needed to understand this step.
Theory: envelope, spatial weighting, harmonics	Define the spatially varying envelope before using its harmonics. Retain the side-by-side map table, product/minimum relation, two-site example, fixed-pair current-ratio law, signed projections, and recurrent weighting. Keep the second-harmonic predictions, amplitude laws, and gain correction here; move the full harmonic ladder and mixed-detector inversion to supplementary Detection.	The reader can distinguish an envelope map from a detector output, then see how amplitude and harmonic measurements separate their predictions.

Argument and retention map

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Place	Scientific content and destination	Contribution to the argument
Theory: steady source	Retain the distinction between nonnegative mean-square polarization, signed source, and circuit response. Retain the convex/concave sign, polarity invariance, beat/DC inequality, 202-fold example, the quadratic subharmonic identity and all uniformity conditions. The envelope-norm proof and orientation averages move to supplementary Spatial maps. The direct tDCS/tACS comparison becomes a separate Theory subsection with the 2–5% result and its amplitude conventions.	The spatial restriction concerns the mean map under stated conditions; it does not prohibit an interior beat maximum. The static-curvature sign rule is distinguished from the dynamic coefficient. The scalar boundary theorem retains its assumptions; general multicompartment bilinearity is stated separately.
Theory: gain, access, operating point	Retain the synaptic attenuation estimates, resonant susceptibility, access–detection–gain product, static versus dynamic coefficients, access-time table, cable estimate, measured versus assumed times, and PEIX identities. Add the conditional kHz source-strength figure, with assumed coupling and explicit units.	After detection is understood, the reader sees what suppresses the carriers and amplifies the beat, and why this does not establish physiological access.
Theory: exact mean field	Move the QIF introduction after the static argument. Keep the MPR equations in physical time and rate, its nonlinear equilibrium curve, threshold recruitment, and slow-synapse and carrier-relaxation conditions. Retain the signs and asymptotes of dynamic mixing.	QIF now answers a question already posed: when does a static curve suffice? A compact crossover derivation and exact coefficient belong in the article model appendix. The fixed-voltage coefficient and exponent catalogue remain in supplementary Detection. The static-rate relation and timescale conditions stay in Theory; the repeated synaptic display moves to the article appendix.

Argument and retention map

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Methods	State the purpose of waveform matching, lock-in, FEM geometry, and each model comparison before settings. Put projection choices with the head model. Place the shortened circuit schematic beside the first model explanation in Theory. Keep the full PING equations, effective input, observables, and convergence summaries. Move run catalogues and extended FEM procedures to topic-organized supplementary sections.	Methods connects each test to its observable. Detailed JR run settings, PING endpoint stability, and QIF cutoff results have explicit supplementary homes.
Results: JR and FEM	Combine the two-sentence beat demonstration with resonance. Retain the two-tone spectrum, curvature, square-law and linearization controls, kHz access checks, FEM maps and the sensitivity conclusion, pinned coupling sweep, timing ratios, and TI/tACS controls.	The evidence proceeds from beat generation to spatial consequences and recurrent amplification. Simplify the FEM figure without removing its maps or numerical comparisons. Call moving foci ROI centroids and keep modest target-to-reference contrasts explicit.
Results: NMM2 and spikes	Retain dynamic-source accuracy, the two-mode coupling curve, gain exponent and finite-amplitude deviations, attracting endpoint, spike cutoff qualification, both field amplitudes, equal-RMS carrier control, sidebands, and phase statistics.	The exact model establishes mixing and gain. A theta-neuron pilot tests cutoff sensitivity at fixed model and readout; time-step, size, window and spike-crossing checks decide the quantitative scope. If unresolved, retain qualitative network evidence. Equal-RMS controls are labeled by the tested model.

Argument and retention map

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Discussion	Group interpretation into access/mixing, spatial predictions, field-induced oscillations and gain, efficiency, population-model limits, and tests. Keep the Violante biological interpretation and conditions, removing repeated displacement values and relocating the ratio algebra. Move the Vieira numerical footnote to supplementary Efficiency while retaining its unmatched-input qualification. Keep the Cifra scope attached to its claim.	Each subsection answers one scientific question. Missey constrains carrier dependence without measuring source transfer. Label the Violante interpretation a hypothesis. Consolidate repeated mechanism explanations; retain the efficiency bound within its family.
Predictions and Conclusion	Merge the nine test rows into six concise claim-to-test entries while retaining independent amplitudes, steering, state/carrier dependence, harmonic calibration, recurrence, and equal-RMS controls. Relocate the full evidence matrix to supplementary Experimental tests. Keep the off-resonance design here.	The conclusion states the supported findings and the measurements that separate mechanisms; detailed qualifications remain beside their claims.
Notation and supplement	Replace tilde variables with explicit physical units in the article and an explicit membrane-time convention in the uncoupled derivation. Keep all equations, citations, supplementary figures, LaNMM results, detector-bound derivation, and vocabulary metrics.	Article appendices contain notation, the JR specification, the slow-synapse display, and compact MPR crossover derivation. Supplementary sections contain access, detection, spatial maps, amplification, efficiency, laminar extension, and metrics/settings; figures accompany their topics. Article-appendix moves do not reduce the article count.

Scope, length, and verification

Approved work proceeds before the next author reading

The main article retains the full PING system, core MPR equations and static-rate curve, Table 2 access evidence, amplitude-matching conventions, and the compact crossover derivation. The feasibility figure replaces scattered access-budget explanations. The theta-neuron pilot passed step and window checks but changed by 12% under population doubling, exceeding its 5% criterion. Its scope is therefore qualitative; the equal-RMS gamma control remains a mean-field result.

The revised abstract distinguishes static and dynamic detection, names the single-head illustration, and describes JR population-rate modulation. The steady effect at depth, 2–5% matched-polarization comparison, shared tACS gain, and scoped boundary restriction remain. The multicompartment statement preserves bilinear current scaling without assuming a universal common projection vector.

The article falls from 14,522 to about 13,000 words including captions and article appendices, with the new results included. Repeated explanations are consolidated and the full prediction/evidence matrix is supplementary. The article remains above the journal's 12,000-word norm. No integral model is relocated merely to improve that count. The LaNMM study remains supplementary. The efficiency-bound placement remains unchanged.

A NumPy-only FEM consumer and checked exporter are prepared for derived fields, coordinates, tissue and ROI labels, orientations, units, and hashes. The IXI002 per-node inputs are unavailable locally; export and real-data comparison remain pending before release. Validation covers analytic arithmetic, numerical convergence, retained claims, article/supplement builds, references, rendered pages, and a redline against both the editing and reading baselines. Klaus checks the integrated revision. Public release and further coauthor circulation remain held.